

Confounding and Interaction

Two of epidemiology's trickiest ideas, made concrete: when a comparison is misleading because the groups were never truly comparable, and when an exposure's effect depends on who you ask.

ROADMAP

Seven Stops

- 01** What confounding is, and a classic example that makes the mechanism concrete
- 02** The three conditions that make a variable a confounder
- 03** Controlling confounding — in the design of a study, and in the analysis
- 04** Assessing confounding when you cannot measure it directly, and how it relates to other biases
- 05** Interaction — an unusually ambiguous word, unpacked with one worked example
- 06** Analyzing effect modification and joint effects in practice
- 07** Bringing confounding and interaction back together

THE CORE IDEA

Confounding Is a Failure of Comparability

Every study rests on a comparison: the exposed group versus the unexposed group. Confounding occurs when those two groups would have had different disease risks even if they had received the same exposure — that is, they were never truly comparable, or “exchangeable,” to begin with.

A useful test: imagine the counterfactual world where the exposed group had NOT been exposed. If, in that imagined world, they still would have had a different disease frequency than the unexposed group actually has, the comparison is confounded — the difference we see is partly just a difference between the groups themselves, not a true effect of exposure.

Exactly what “exchangeable” must mean depends on which causal effect is being targeted. To estimate the effect among the exposed specifically — the most common target — the unexposed need only represent the baseline risk the exposed group would have had without exposure: a narrower, more attainable requirement than full, two-way exchangeability.

In everyday terms: confounding is a mixing-up of the exposure’s own effect with the effect of some other factor that happens to travel along with it.

A CLASSIC ILLUSTRATION

Beta Carotene, Smoking, and Lung Cancer

Suppose we compare the risk of lung cancer in people with low versus high dietary beta carotene intake. The two groups are very unlikely to be comparable: people who eat less beta carotene tend to be less health-conscious overall, and so are also more likely to smoke.

If smoking is more common in the low-beta-carotene group, smoking alone will raise that group's lung-cancer risk — even if beta carotene itself has no effect on lung cancer whatsoever.

The result: a spurious inverse association between beta carotene and lung cancer, created entirely by smoking — especially if adjustment only uses crude categories like non-smoker, ex-smoker, or current smoker.

Any variable that affects disease risk in the absence of exposure has the potential to confound this way — and smoking, in this example, does exactly that.

THE CORE IDEA

Which Way Does the Bias Push?

Higher-risk levels of the confounder are more common among the exposed

The bias pushes the estimated effect upward.

Higher-risk levels of the confounder are more common among the unexposed

The bias pushes the estimated effect downward.

In extreme cases, confounding can even reverse the apparent direction of an effect — making a harmful exposure look protective, or vice versa. It can also manufacture a spurious association out of nothing, when the exposure under study truly has no effect on disease at all.

THE CORE IDEA

Standing In for a Confounder You Can't Measure

Factors associated with a true confounder can act as confounders themselves, and serve as usable surrogates for it — provided they are not affected by the exposure or the disease being studied.

Socioeconomic status is a common example: it can serve as a surrogate measure for a whole bundle of causal factors — living conditions, lifestyle, access to preventive care — that are themselves difficult or impossible to measure directly, but that plausibly confound many exposure–disease relationships.

CONFOUNDERS

Three Conditions, Necessary but Not Sufficient

A confounder is a variable partly responsible for the exposed and unexposed groups not being exchangeable. Three conditions are traditionally required of it — though meeting all three is still not, by itself, an ironclad guarantee.

1. Predicts disease even when exposure doesn't vary

2. Is associated with the exposure in the source population

3. Is not itself affected by the exposure or the disease

CONDITION ONE

It Has to Predict Disease on Its Own

To confound the effect in the exposed, a factor has to be predictive of disease even when exposure itself does not vary — a “background” or “baseline” risk factor. It need not be a genuine cause of disease; merely being reliably predictive within exposure levels is enough.

Surrogates for causal factors — ethnicity, gender, socioeconomic status — can count as potential confounders under this condition even when they are not themselves direct causes. This is also why age and sex are adjusted for in almost every study of disease: both are known, strong, background predictors of risk.

When prior knowledge is not available, investigators must instead turn to the study’s own data as a guide — a weaker, more error-prone substitute for genuine background knowledge.

CONDITION TWO

It Has to Track the Exposure, in the Source Population

A confounder must be associated with the study exposure in the source population the study was drawn from — not merely inside the sample the study happens to have collected.

In a case-control study, a background risk factor should be associated with exposure among the controls, at least if controls were selected without bias. If it isn't associated with exposure in the source population, but chance sampling makes it look associated among the cases, that is not confounding — the factor's apparent link to exposure never reflected the population it was drawn from.

CONDITION THREE

It Must Not Sit Between Exposure and Disease

A variable affected by the exposure — an intermediate on the causal pathway to disease — or affected by the disease itself, should not be treated as an ordinary confounder. Conventional adjustment for it can bias the total-effect estimate; special causal methods are needed on the rare occasions when an intermediate itself must be part of the analysis. Variables affected by the outcome remain especially problematic either way.

Example: studying obesity's effect on coronary heart disease, it would be wrong to control for hypertension if hypertension is itself a step on the path from obesity to heart disease. (If hypertension were the exposure of primary interest instead, obesity would rightly be treated as a potential confounder of it.)

The same caution applies to controlling for health outcomes that may themselves be part of disease genesis — such as reduced lung function following exposure to a respiratory hazard, in a study of chronic lung disease.

CONFOUNDERS

A Judgment Call, Not a Fact Read Off the Data

Because being a confounder involves judgments about causal and temporal ordering, it cannot be determined from statistical data alone. It requires outside knowledge of what causes what, and what happened before what.

A common shortcut in practice is the “change-in-estimate” approach: control a suspected confounder, and see how much the effect estimate moves. Near-equality of the crude and adjusted estimates is treated as evidence of no important confounding. This is convenient — but it can mislead, particularly when the confounder or the exposure is misclassified, or when the effect measure is an odds ratio or rate ratio.

CONFOUNDERS

A Worked Example: Confounding by Smoking

A hypothetical study of occupational asthma. Two-thirds of the exposed group smoke, versus one-third of the non-exposed — and smoking is itself a background risk factor for asthma.

	Smokers Exposed	Smokers Non-exposed	Non-smokers Exposed	Non-smokers Non-exposed	Total Exposed	Total Non-exposed
Asthma cases	800	400	200	400	1000	800
Non-cases	1200	600	800	1600	2000	2200
Total	2000	1000	1000	2000	3000	3000
Prevalence	40%	40%	20%	20%	33.3%	26.7%
Prevalence ratio	1.0		1.0		1.25	

Within each smoking stratum the prevalence ratio is exactly 1.00 — exposure has no real effect. But the crude ratio, combining both strata, comes out to 1.25. Smoking — more common among the exposed, and a risk factor for asthma regardless of exposure — manufactures the appearance of an effect that isn't there.

A SUBTLE TWIST

The Same Variable Can Confound in Opposite Directions

In studies involving aggregate-level effects — ecological or multilevel studies — a factor at one level of aggregation may confound quite differently than the “same” factor measured at another level.

Both an individual’s income and their neighborhood’s average income can separately predict robbery risk — sometimes in opposite directions. Robbery rates run higher in low-income neighborhoods overall, yet within a given neighborhood, an individual’s own robbery risk could still rise as their personal income rises. Neighborhood income and individual income could both be genuine confounders, yet point the resulting effect estimates in opposite directions.

The lesson: whatever level of interest — country, neighborhood, individual — it is often essential to measure and adjust for variables at the other levels too.

CONTROLLING CONFOUNDING

Confounding Can Be Controlled — Rarely Eliminated

Confounding can be reduced, not usually eliminated completely, either in the design of a study, in the analysis, or both. Three common design-stage methods:

Randomization

Restriction

Matching

DESIGN CONTROL, 1 OF 3

Randomization

Random allocation of participants to exposure categories — but only an option when the exposure is a precise treatment protocol the investigators can actually assign. It would be impractical and unethical, for instance, to randomize people to smoke.

Randomization does not prevent all confounding — the final analysis can still be confounded by the effects of non-adherence and drop-out. What it does provide is a foundation for statistical inference about causal effects from observed associations, since treatment assignment is guaranteed to be unaffected by baseline, pre-randomization factors.

One specific source it does prevent: “confounding by indication,” where a physician’s choice of treatment is itself guided by a patient’s prognosis. And the random assignment itself — the “intent-to-treat” variable — can serve as an instrumental variable to partly address bias from non-compliance or differential loss to follow-up.

DESIGN CONTROL, 2 OF 3

Restriction

Narrow the study to a restricted range of confounder values — for example, restricting a study to white males aged 35 to 54. Confounding by age, sex, or race simply cannot arise if everyone studied shares the same values on those variables.

This has real conceptual and computational advantages, especially when the source population was already mostly limited to a special group, as in many occupational studies. But it can also severely limit the number of potential study subjects, and ultimately the generalizability of the findings. A major objection to historical occupational research is that it often restricted itself unnecessarily to white males, leaving possible differences in effect among other groups unexamined.

DESIGN CONTROL, 3 OF 3

Matching

In a cohort study, matching each exposed participant to an unexposed one of the same age and sex prevents any baseline association between those factors and exposure — but it is expensive and time-consuming, so it is seldom done.

In case-control studies, matching does not prevent confounding — adjustment in the analysis is still required. Worse, matching on a factor only weakly related to disease but strongly related to exposure can introduce a selection bias that must then be corrected, adding analytic complexity and potentially harming precision. Choosing a sibling as a control for childhood cancer, when studying a home exposure, is a classic example: siblings sharing a home means case and control are exposure-concordant almost by construction, leaving little discordant information to estimate from.

Even when it works well, matching's precision gain is often modest — typically just a 5 to 15 percent reduction in the variance of the effect estimate — so the practical cost of finding suitable matches often outweighs the statistical benefit.

A COMMON CONFUSION

Randomization Is Not the Same as “No Confounding”

Randomization means treatment is assigned at random within levels of covariates known at the time of assignment — but this only balances covariates *probabilistically*, across many hypothetical repetitions. It does not force an equal distribution of covariates in any single study.

The traditional, stricter “no confounding” condition requires that, within observed covariate levels, the different treatment groups would have identical risks if given the same treatment — ruling out even random baseline differences. Treating randomization as automatically delivering “no unmeasured confounding” or full “ignorability” is a common but genuinely fallacious move, on both statistical and causal grounds.

RANDOMIZED TRIALS

Why Adjust for Baseline Even After Randomizing?

Statistically, adjusting for a measured baseline imbalance improves efficiency — it reduces the variance of the estimated average effect. Causally, failing to control for an observed baseline imbalance is like ignoring evidence of “random confounding” that happened to arise from the randomization itself, and which adjustment can reduce.

And randomization only ever applies to the baseline assignment of exposure. Even a properly randomized trial remains vulnerable to bias from events that happen afterward and outside experimental control — heavy post-randomization loss to follow-up, selective drop-out linked to treatment side effects, and non-compliance.

CONTROL IN THE ANALYSIS

Adjusting After the Data Are In

Confounding can, in theory, be controlled in the analysis by adjusting simultaneously for all confounding factors, or a sufficient subset of them — typically through stratification or regression modeling.

This presumes the confounders were measured accurately, which is often not the case, and that there is enough data to adjust for all of them, which is also often not the case. When a nonlinear model — logistic or proportional-hazards regression — is used with too many covariates entered at once, the estimated coefficients can become seriously inflated.

CONTROL IN THE ANALYSIS

Stratification Runs Out of Room Fast

In its simplest form, controlling confounding in the analysis means stratifying the data by the confounder's levels and calculating a summary effect estimate — a weighted average across the strata. Grouping age into 5 categories and sex into 2 already creates 10 confounder strata to manage at once.

It is usually not possible to control simultaneously for more than two or three confounders this way, since finer stratification quickly produces strata containing no exposed or no non-exposed people — strata too sparse to be informative. Regression modeling mitigates this to some extent, by “smoothing” the data across confounder strata rather than partitioning it into ever-thinner slices.

A COMMON TRAP

The “Table 2 Fallacy”

When a regression model reports effect estimates for more than one variable, it is tempting to read every listed coefficient as that variable’s own total effect. That temptation is the trap.

Suppose a model estimates both age’s and vaccination status’s effects on mortality. Age is a strong confounder of the vaccination effect, so it rightly belongs in the model for that purpose. But that same model, at best, only estimates age’s “direct” effect — net of whatever age’s effect on mortality runs through vaccination status. To validly estimate age’s *total* effect, vaccination status would need to be dropped from the model entirely, since it is an intermediate variable on age’s own causal pathway, not a confounder of it.

The covariates a model needs to estimate one exposure’s effect are rarely the right covariates for reading off another variable’s effect from the same model.

ASSESSING CONFOUNDING

When You Lack Data on a Suspected Confounder

Even without direct data, it is often still possible to assess the likely direction and magnitude of confounding — through surrogates, related diseases, and a growing toolkit of more formal methods.

Occupational studies of lung cancer are a good example. Smoking is the strongest common risk factor for lung cancer, with a ten-fold or greater relative risk — yet smoking information is rarely available, or only available crudely (smoker/non-smoker). In historical data, smoking rarely confounded these studies by more than 1.5-fold, because smoking prevalence tended not to vary dramatically across the occupations being compared.

ASSESSING CONFOUNDING

A Trick: Check a Different Disease

One approach to gauging unmeasured confounding is to analyze a confounder-related disease other than the one actually under study.

If a researcher worries that unmeasured drinking habits might confound a study of an alcohol-related cancer, checking liver cirrhosis mortality — a disease closely tied to heavy drinking — across the compared groups can help. If cirrhosis mortality does not differ meaningfully between the groups, that suggests drinking prevalence probably doesn't differ much between them either, making a large confounding effect from unmeasured drinking less likely.

ASSESSING CONFOUNDING

More Formal Tools for Unmeasured Confounders

A growing family of design and analysis techniques targets unmeasured confounding directly, under specific assumptions:

Negative-control designs

Part of a broader approach known as triangulation

Difference-in-differences

Comparing changes over time across groups

Proximal causal inference

Using measured proxies for the unmeasured confounder

Instrumental variables

A baseline variable related to the outcome only through exposure

Each of these methods selects or constructs comparison groups or adjustment variables that permit relatively unbiased effect estimation — but only under assumptions of their own that cannot always be guaranteed to hold.

ASSESSING CONFOUNDING

Sensitivity and Bias Analysis

Rather than assuming confounding away, a sensitivity or bias analysis directly estimates the potential direction and extent of confounding — using external, contextual (“prior”) information about how prevalent an unmeasured confounder likely is, and how strongly it likely relates to the outcome.

Given enough such information, a full probability distribution for the confounder-adjusted association can even be derived. But that distribution can be seriously misleading if the input assumptions are not firmly grounded — which is why, when the validity of underlying assumptions is uncertain, more cautious language like “compatibility interval” is often preferred over the more confident-sounding “confidence interval.”

ASSESSING CONFOUNDING

The E-Value: A Popular Shortcut

One widely offered shortcut around the labor of a full sensitivity analysis is the E-value: the smallest strength of association — on the risk-ratio scale — that an uncontrolled confounder would need to have with *both* the exposure and the outcome, in order to completely explain away an observed exposure-outcome association.

A larger E-value is often read as reassurance that a modest, plausible confounder couldn't account for the finding. But the E-value has been criticized as insufficient on its own: it only characterizes results under one particular, extreme confounder-prevalence scenario, rather than surveying the plausible range — so it is best treated as a quick first check, not a substitute for a proper bias analysis.

A TOOL WORTH NAMING

Causal Diagrams for Identifying Confounders

Causal directed acyclic graphs, or DAGs, give a systematic, visual way to separate genuine confounders from non-confounding covariates, and to guide both study design and analysis strategy. They now underpin a large share of modern confounder-identification practice — a rich enough subject to deserve a chapter of its own.

THREE BIASES, NOT ONE

Confounding Is Not Selection Bias

Confounding is non-exchangeability of the exposed and non-exposed subgroups within the source population — a property of that source population itself. Selection bias is something else entirely: bias arising from how participants are selected, or select themselves, out of that source population.

Selection bias is not usually an issue in a cohort study or randomized trial with complete recruitment and follow-up, because in that case the study group simply is the entire source population. It becomes a much bigger concern in case-control studies, where the control group must somehow represent a source population it was never guaranteed to fully capture.

SELECTION BIAS

Three Ways It Sneaks In

Choosing controls with other diseases. Only safe if the exposure being studied doesn't meaningfully affect the risk of a large share of those other diseases — otherwise the control group misrepresents the source population's true exposure pattern.

Non-response at a cohort's baseline. This can itself become a source of confounding, since non-response can create associations between exposure and other risk factors that would not otherwise exist.

Matching on a factor affected by exposure. Even a factor that is not itself a disease risk factor can be turned into a confounder that must be controlled, if matching on it creates a spurious factor-disease association.

When selection is driven by exposure and by case-control status together, stratifying on the selection-related factor afterward will rarely produce valid estimates.

THE FULL PICTURE

Information Bias, and Sorting the Biases Out

Information bias is the result of misclassifying study participants' exposure or disease status — distinct from selection bias (who gets into the study) and confounding (non-comparability of those who did).

Some treat any bias that analysis alone could control as “confounding” — but that definition is too broad, since controlling for anything in an analysis still requires background information to do correctly. The clearer, traditional taxonomy: confounding is a mixing or confusion of the effects of other factors with the exposure's own effect — a concept traceable at least to John Stuart Mill — while other biases arise from biased selection of subjects, mismeasurement, or the misapplication of statistical methods, such as controlling for a mediator or a collider, or using large-sample statistical methods on data too sparse for them to work correctly.

Most real studies suffer from more than one bias at once — and the combined effect of several biases together can be genuinely counterintuitive.

INTERACTION AND EFFECT MODIFICATION

An Unusually Ambiguous Word

In statistics, “interaction” usually just means the need for a product term in a regression model. In epidemiology, it usually means something more substantive: that to estimate exposure’s effect on an outcome, we must first know the level of some other factor.

Biological interdependence

Synergism and antagonism between mechanisms in an individual

Effect-measure modification

Heterogeneity of an effect measure across a population

A KEY DISTINCTION

An Effect Modifier Need Not Be a Confounder

Confounding and interaction are quite distinct concepts. A variable can modify an effect without confounding it, confound it without modifying it, do both, or do neither.

If the percentage of smokers is identical in the exposed and non-exposed groups, smoking cannot confound the exposure's effect — there is no exposure–smoking association to create bias. Yet the rate ratio for the exposure's effect could still differ sharply by smoking status: doubling the disease rate in smokers, say, while having no effect at all in non-smokers. In that situation, smoking is not a confounder — but it is very much a modifier of the rate ratio being used to measure the exposure's effect.

A WORKED EXAMPLE

The Same Overall Numbers, a Different Story

Same overall asthma prevalence ratio as before (1.25) — but now smokers make up exactly half of both the exposed and non-exposed groups, so there is no confounding by smoking at all.

	Smokers Exposed	Smokers Non-exposed	Non-smokers Exposed	Non-smokers Non-exposed	Total Exposed	Total Non-exposed
Asthma cases	600	400	400	400	1000	800
Non-cases	900	1100	1100	1100	2000	2200
Total	1500	1500	1500	1500	3000	3000
Prevalence	40%	26.7%	26.7%	26.7%	33.3%	26.7%
Prevalence ratio	1.5		1.0		1.25	

The prevalence ratio is 1.5 among smokers but only 1.0 among non-smokers — real effect-measure modification. Assessing confounding compares the crude estimate to the adjusted one; assessing modification compares the stratum-specific estimates to each other. Here the crude and adjusted estimates roughly agree, yet the two strata still tell very different stories.

A REAL STUDY

Asbestos, Smoking, and Lung Cancer

Lung-cancer mortality rates per 100,000 person-years, comparing asbestos workers to workers in other blue-collar occupations, by smoking status (Selikoff et al. 1980, via Steenland & Thun 1986).

	Rate in Smokers (RR)	Rate in Non-smokers (RR)	Rate Ratio
Asbestos	935.8 (RR11 = 32.7)	500.5 (RR01 = 17.5)	1.9
Non-asbestos	199.5 (RR10 = 7.0)	28.6 (RR00 = 1.0)	7.0
Rate ratio	4.7	17.5	
Rate difference	736.3	471.9	

All rates adjusted for age and calendar period. The bottom “Rate ratio” and “Rate difference” rows give asbestos’s effect measured separately within each smoking stratum.

THE AHA MOMENT

One Dataset, Two Contradictory Stories

On the rate **DIFFERENCE** scale

Asbestos's effect is bigger in smokers: 736 per 100,000 versus 472 in non-smokers.

On the rate **RATIO** scale

Asbestos's effect is bigger in non-smokers: a 17.5-fold increase, versus 4.7-fold in smokers.

Both statements describe the very same underlying numbers. Which group asbestos “affects more” depends entirely on which effect measure is asked. This is exactly why assessing interaction has long been called “model-dependent”: whenever an exposure has an effect at all, every other causal risk factor will modify either the rate ratio or the rate difference — uniformity on one measure implies non-uniformity on the other.

TWO KINDS OF INTERACTION

Mechanisms in a Person vs. Patterns in a Population

Biological interaction

Two factors are “synergistic” if both must be present, and “antagonistic” if one blocks the other, for a specific disease mechanism to operate in a given individual.

Effect-measure modification

The effect measure, calculated across a population, simply varies by the level of some other factor — whatever the underlying biology.

A closely related, “potential-outcome” definition: biological interaction is present when an individual’s pattern of potential outcomes to one factor changes as the “interacting” factor changes. Under this view, non-additivity of individual risks can, under further assumptions, signal the presence of biological interaction — but the reverse is not guaranteed.

A DEEP LIMITATION

Biological Interaction Cannot Be Read Off the Data

Biological interaction concerns interdependence between mechanisms of action — a fact about individuals. Effect-measure modification concerns population patterns. Because of that gap, biological interaction is not fully identifiable, or estimable, from epidemiological data alone, even when everything is measured and unbiased.

Even ideal additivity of risks would not definitively rule out biological interaction: two factors could still be joint components of the same underlying disease mechanism while having competing or antagonistic effects that happen to balance out and look additive. And two factors can be interdependent components of a shared disease process without ever acting at the same place or the same time in the body — for instance, if one affects an early stage of a multistage process and the other affects a much later stage.

A MODELING CHOICE

What Counts as “No Interaction”?

The sufficient-cause and potential-outcome definitions of biological interaction are attractive because they rest on an explicit causal model — and under simplifying assumptions, they point toward additive-risk models as the natural scale for “no interaction.”

But many different underlying biological models can produce exactly the same observed pattern of risks or rates across compared populations — so the specific mechanism responsible can never be identified from epidemiological data alone, no matter which scale, additive or multiplicative, the analysis happens to use.

A MODELING CHOICE

Why Multiplicative Models Are the Practical Default

Better statistical behavior — additive models can face convergence problems, and their inference can be unreliable except in very large studies

Real joint effects often look closer to multiplicative than additive, so fewer interaction terms are usually needed

Multiplicative models never predict an impossible negative risk or rate; additive ones can

Multiplicative-model summaries tend to sit closer to a true population-average effect than additive-model summaries do

In typical case-control studies, confounder-adjusted assessment of risk additivity is difficult or sometimes impossible, while departures from multiplicativity are much easier to assess

Departures from multiplicativity, unlike departures from additivity, can be assessed in the same straightforward way from either cohort or case-control data — one more reason ratio scales dominate applied practice.

ANALYZING EFFECT MODIFICATION

Estimate by Stratum, Compare, and Be Careful

The usual approach: estimate the effect measure for the exposure separately within each level of a presumed modifier, then compare those stratum-specific estimates to each other.

This approach lacks precision and power, so it can be quite misleading to conclude “no modification” just because a formal test produced a large p-value — the same fallacy as concluding “no effect” from a large main-effect p-value. With several candidate modifiers, examine each one separately, or model continuous modification directly; prior selection of the modifiers of greatest genuine interest helps keep the task manageable.

ANALYZING EFFECT MODIFICATION

Heterogeneity Doesn't Sink the Overall Estimate

Some sources hold that an overall effect estimate shouldn't even be calculated once modification is present. In practice this principle is often set aside when the stratum-specific estimates aren't too different — and measures like standardized risk ratios and IPTW-style estimators can still validly estimate a meaningful average causal effect even under real heterogeneity.

A separate caution: don't go fishing. Testing every possible two-way and three-way product term in a preliminary model, then keeping whichever ones clear some p-value cutoff, tends to produce complex, unstable models plagued by data sparsity. It is usually preferable to deal with confounding first, establish whether an overall effect exists, and only then investigate effect modification deliberately — guided by genuine prior interest, not a blind search.

A SHARED OBSTACLE

Measurement Error Muddies Both Questions

Measurement errors — misclassification of exposure, disease, confounder, or modifier variables — are a major obstacle to assessing effect-measure modification as well as confounding.

Such errors can make an effect measure appear to vary across strata when in reality it does not, or make it appear nearly constant when in reality it does vary. In an analogous fashion, other biases that differ across strata — such as residual confounding or selection bias — can spuriously create or mask apparent modification too. Sensitivity analyses can in principle explore this, but in practice there are often too many uncertain parameters to form a genuinely clear picture.

ASSESSING JOINT EFFECTS

The Practical Endgame: Present All Three Comparisons

When two factors have a genuine, a priori reason for interest — not just any pair of covariates — the recommended presentation shows relative risks for factor A alone, factor B alone, and both together, each compared to persons exposed to neither.

These should be adjusted for genuine confounders, but not for the two factors of interest treated as modifiers of one another. The needed table can be derived from any model form — a multiplicative model with product terms, or simply separate indicator variables for each mutually exclusive combination of exposure.

BACK TO ASBESTOS AND SMOKING

Is the Joint Effect Bigger Than Additive?

The two single-exposure rate ratios, each compared to workers exposed to neither factor, were 7.0 and 17.5. If the two exposures simply added their effects, the joint rate ratio would be:

$1 + (7.0 - 1) + (17.5 - 1) = 23.5$, if purely additive

32.7, the actual observed joint rate ratio

The observed joint effect is clearly more than additive — a positive departure from additivity, consistent with (though not proof of) the two exposures acting as joint components of a shared disease mechanism.

AN IMPORTANT CAUTION

What Non-Additivity Does — and Doesn't — Prove

Non-additivity like this is consistent with asbestos and smoking being joint components of at least one shared sufficient cause. One could even argue the observed non-additivity refutes the idea that they never interact biologically at all — assuming, as always, no residual confounding or bias.

But the reverse case is more subtle than it looks: even if the joint effect HAD come out exactly additive, that would only have favored — not proven — the idea that the two factors are not joint components of a common sufficient cause. Two factors could still share a sufficient cause while also having competing or antagonistic effects that happen to balance out and produce apparent additivity. Even in ideal, unbiased circumstances, additivity does not definitively rule out biological interaction in some individuals.

SCREENING MANY EFFECTS

The Multiple Comparisons Controversy

Screening a large number of joint effects or subgroup estimates raises a real statistical hazard: some estimates in any large collection will look large purely from random error, and it is tempting to chase whichever ones stand out.

Classical multiple-comparisons corrections, such as Bonferroni-style adjustments, are controversial because they only guard against false positives — while dramatically inflating false negatives in the process. If false negatives matter more to the question at hand than false positives, a classical correction can end up performing worse than making no adjustment at all.

A BETTER FIX

Hierarchical Modeling Beats Both Extremes

Hierarchical or multilevel modeling — also known as penalized estimation, random-coefficient regression, shrinkage estimation, or empirical-Bayes and semi-Bayes regression — is demonstrably superior to both extremes: no adjustment at all, and classical multiple-comparisons correction.

This superiority holds up across theory, simulation, and real epidemiological examples. The same family of methods can also be used for confounder control, as an alternative to formal variable-selection procedures, and is implementable in standard statistical software.

No analytical method, however, can overcome the fundamental limits of precision in typical epidemiologic data — in practice, usually only fairly substantial effect modification or interaction will actually be detectable.

SUMMARY AND CONCLUSIONS

Two Ideas, Cleanly Restated

Confounding

The exposed and non-exposed groups would have had different outcomes even if both had received the same exposure — they were never truly exchangeable.

Interaction

A highly ambiguous term, subsuming two separate ideas: estimable effect-measure modification, and biological interdependence that data alone cannot fully reveal.

SUMMARY AND CONCLUSIONS

Same Covariate, Two Different Questions

Assessing confounding

Compare the crude, unadjusted estimate to the adjusted one.

Assessing effect-measure modification

Compare the stratum-specific estimates to each other.

The same covariate — smoking, in every example here — can be a confounder, an effect-measure modifier, both at once, or neither, depending entirely on how it happens to relate to exposure and to disease in a given study. Effect-measure modification is algebraically related to, but not causally equivalent to, the statistical “interaction” of most textbooks: it specifically concerns the exposure’s actual effect, not merely a pattern in possibly-confounded association.

BOTTOM LINE

Comparable Groups, Honestly Measured

Confounding is what happens when the groups being compared were never truly comparable to begin with. Interaction, in its most useful epidemiological sense, is the reminder that an exposure rarely has just one effect — its apparent size depends on who you look at, and which measure you choose to ask the question with.

Neither problem has a mechanical fix. Both call for the same thing: outside knowledge of the causal story, honestly applied, before the numbers are ever allowed to speak for themselves.

From a hypothetical asthma survey to a cohort of asbestos workers, the same two questions recur in every comparison epidemiology ever makes.

SOURCE & ATTRIBUTION

Pearce, N., & Greenland, S. (2024). “Confounding and Interaction.”

In: *Handbook of Epidemiology*, W. Ahrens & I. Pigeot (eds), Springer Nature, 3rd ed.

Prepared by Omokhodion Alfred Eriakha · September 21, 2026

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